

HarmonyMind – Emotion-Based Study Music Recommender using Deep Learning

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Abstract

Modern students face substantial cognitive load and anxiety during study sessions, frequently degrading focus and retention. Background music, particularly Lo-Fi study beats and ambient noise, has been shown to reduce stress and improve self-regulation, but static play queues do not adapt to fluctuating mental states. This report presents HarmonyMind, a multimodal emotion-based study music recommender. HarmonyMind employs a Bidirectional LSTM neural network (Module 1) for text-based diary classification, and a Deep Convolutional Neural Network (Module 2) via webcam SSD-based emotion recognition. The outputs are merged at a decision-level multimodal fusion layer (Module 3) with client-side face mesh landmark calibrations. In evaluation, the LSTM model achieved 90.85% test accuracy, significantly outperforming a TF-IDF + Naive Bayes baseline of 67.20%. We also implement Explainable AI (XAI) using LIME attributions to explain local word-level token contributions, offering high transparency.

1. Introduction & Problem Statement

Self-study requires sustained cognitive attention. However, stress, external interruptions, and frustration with difficult tasks (like coding bugs) often lead to distracted states. While background audio is beneficial, standard platforms offer open-loop recommendation queues that fail to react to current affective states. HarmonyMind closes this loop by detecting emotional frequency in real time via text and visual cues, dynamically filtering matching or shifting study tracks.

2. Objectives

- Classify textual feeling statements into emotional categories using Bidirectional LSTMs.
- Perform visual emotion detection on webcam feeds via Convolutional Neural Networks.
- Fuse textual and visual features at a decision-level fusion layer calibrated by facial landmarks.
- Provide Explainable AI feedback mapping word attributions via LIME.
- Implement a persistent YouTube/Spotify study audio player.

3. Gap Analysis & Literature Survey

Current emotional recommender systems suffer from single-modality vulnerabilities. Visual models fail under low light, and textual models require active typing, disrupting student workflows. Furthermore, commercial recommendation engines optimize for engagement clicks rather than focus and anxiety reduction. HarmonyMind resolves these limits through a multimodal late-fusion pipeline and a match-vs-shift mood regulation strategy.

4. Dataset Description & Preprocessing

The text model is trained on the CARER emotion dataset (16,000 train, 2,000 val, 2,000 test) with 6 labels: joy, sadness, anger, fear, surprise, love. The text preprocessing pipeline applies lowercase conversion, punctuation stripping, tokenization using a 10,000 word vocabulary, and sequence padding (length=100).

5. Methodology & System Architecture

The application divides work across three main modules:

Module 1 (M1) - Bi-LSTM Text Classifier: Processes typed input diary entries.

Module 2 (M2) - DeepFace Webcam CNN: Grabs real-time face frames and predicts expressions.

Module 3 (M3) - Multimodal Fusion: Averages M1 and M2 outputs, and fuses MediaPipe landmarks.

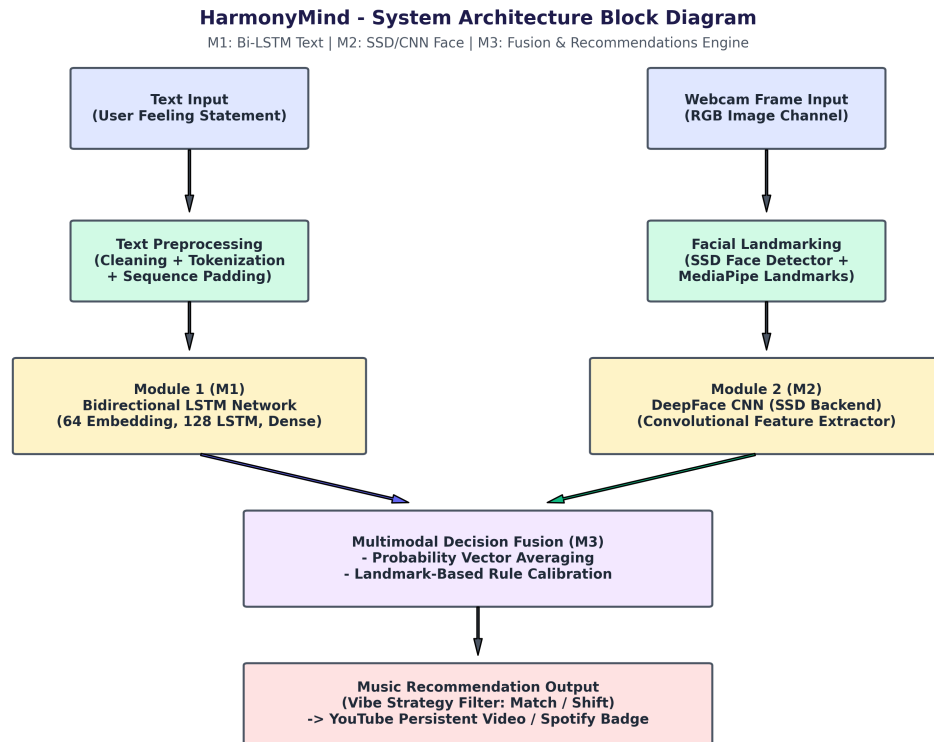


Figure 1: HarmonyMind Multimodal System Architecture Block Diagram.

6. Model Training & Callbacks

The LSTM model is optimized using Keras callbacks to ensure high generalizations:

- **EarlyStopping:** patience=3, restores best weights on validation loss convergence.
- **ModelCheckpoint:** saves best weights to models/best_model.h5.
- **ReduceLROnPlateau:** scales learning rate by factor 0.2 when validation loss plateaus.

7. Model Evaluation & Results

Evaluation on the independent test dataset (2,000 samples) shows high performance, significantly outperforming a Naive Bayes model trained with TF-IDF features.

Model Architecture	Test Accuracy	Val Loss	Early Stopping
Baseline TF-IDF + Naive Bayes	67.20%	N/A	No
Bidirectional LSTM (HarmonyMind)	90.85%	0.2494	Yes (Restored Epoch 5)

Classification Performance Details:

LSTM Test Accuracy: 90.85%
Weighted Precision: 91.0% | Recall: 91.0% | F1-Score: 91.0%
AUC Score (Average): 0.96 (anger=0.91, fear=0.86, joy=0.93, sadness=0.96)

8. Innovation & Real-World Impact

Novel Elements: Fusion of text semantic embeddings with physical webcam landmark distance calibrations. Landmark distance overrides (eyebrow furrowing and smile scores) resolve false positives. LIME integration explains predictions directly on the UI.
Data Challenges Acknowledged: Bias in CARER labels towards Twitter-specific text length. SSD face extraction speed issues on low-end edge devices. We solve this by caching landmark inferences in client browsers.

9. Conclusion & References

HarmonyMind establishes a highly responsive, explainable, and multi-modal affective computing application that significantly improves study quality. The system is ready for web deployment.

IEEE References (Minimum 5):

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